

An Overview on Copula Function Methods in Credit Portfolio Modeling

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Summary

We provide an overview on copula function methods in credit portfolio modeling. The rationale of using copula function and the study of correlation from conditional perspective are given. We also present all technical details on the valuation of small basket derivatives of *i*th-to-default, and CDOs. Fast algorithms on the valuation are detailed. We briefly discuss about the various mapping algorithms used in practice from the index tranche base correlation to the pricing of bespoke portfolio. Risk measures and hedging issues are discussed. Further research should try to bring more theory into the framework, and more empirical tests on the efficacy of hedging based on the current model as well as any new alternative model.

Introduction

The rapidly growing credit derivative market has created a new set of financial instruments, which can be used to manage the most important dimension of financial risk - credit risk. In addition to the standard credit derivative products, such as credit default swaps and total return swaps based upon a single underlying credit risk, many new products are now associated with a portfolio of credit risks. A typical example is the product with payment contingent upon the time and identity of the first or second-to-default in a given credit risk portfolio. A common variation includes instruments with payment contingent upon the cumulative loss of a given portfolio before a given time in the future. Tranches from collateralized debt obligations (CDOs) are good examples since the investors of the tranche would make payment when the loss of the portfolio exceeds a certain level, but the payment is capped by the tranche size. It is similar to an insurance contract whose payment had a deductible and a ceiling. As more financial firms try to manage their credit risk at portfolio level, and investors become more comfortable with the risk/return profile of the structures, basket credit derivatives and the CBO/CLO market will continue to

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expand. How to model credit risk at portfolio level will become more and more important.

For an investor of fixed income securities such as a life insurance company or a mutual fund it could have credit exposures to hundreds of corporations. After having done enough due diligence to assess each individual credit separately it has to be able to assess its credit risk at the portfolio level. While the portfolio modeling technique of stocks due to Harry Markowitz is well understood by most practitioners, little has been written on how to model a portfolio of fixed income securities subject to defaults. The major difficulties associated with a credit portfolio modeling are as follows. First, we have to model each individual credit accurately since the portfolio is usually not as large as some other credit sensitive portfolio such as credit card portfolio where we could use the classification technique and then use some aggregation techniques in insurance modeling to model each classified class instead of each credit. It also contains more than a few underlying credits which excludes exhaustive scenario analysis to cover all possible outcomes associated with the number of defaults and the timing of default without using Monte Carlo simulation. Second, there exists a thorny issue on how to model default correlation. In practice there might have no previous example of one company's default that triggered another company to default. But empirical studies show that defaults occur more in a bulk than purely randomly. For example recent annual default rates of US corporations are relatively lower comparing to previous annual default rates right after the Internet boom era when default rate was relatively higher, especially in the telephone and telecommunication industry due to the overcapacity. Default, in general, is a "low frequency, but high severity" event, which requires special techniques to study. Default correlation of rare events is even harder to study, but it is too important to ignore. A credit portfolio consisting of independent credits has different properties than one of correlated credits. Third, we are usually concerned with the credit problem over a longer time horizon more than one year. But most of the finance models are built over a single time horizon with a short time span. In the study of default the factor of time is an important factor that we focus on. Fourth, correlation affects liabilities written against a portfolio in an uneven fashion. For example, correlation is the most important parameter in the analysis of risk of the most senior tranche of CDO while some mezzanine tranches of CDOs could be almost insensitive to correlation changes.

In a previous paper Li (2000) has introduced a copula function approach to credit portfolio modeling. In this approach the author first introduces a random variable to denote the survival time for each credit and characterizes its property using a density function or a hazard rate function. This allows us to totally move away from a one period framework so that we could incorporate the whole term structures of default probabilities. Then, the author introduces the use of copula functions to combine information from all individual credits, and in the mean time, to assume a correlation structure among all credits. Mathematically the concept of copula functions allows us to construct a joint distribution of survival times with given marginal distributions as specified by individual credit curves. This two-stage approach to forming a joint distribution of survival times has its advantage. First, it incorporates all information on each individual credit. Second we have a vast choice of different copula functions to form a good joint distribution to serve our purpose than if we assume a joint

distribution of survival times from the start. While the normal copula function is used by the author for illustration due to the simplicity of its economic interpretation of the correlation parameters and relative easy computation of its distribution function, the framework does allow using any other copula functions. Finally the author also introduces an efficient “one step” simulation algorithm of survival times in the copula framework by exploring the mathematical property of copula functions in contrast to the period by period simulation as suggested earlier by others.

At the early stage of using copula function, the market was using different correlations for different tranches. For example, senior tranche tended to imply a much higher correlation parameter in order to match the market traded spread for the tranche. The correlation associated with each tranche is called a *compound correlation*. But this compound correlation approach of pricing each tranche using different a flat correlation has its problem. For example, the total loss of all tranches from a whole capital structure CDO might not add up to the total loss of the underlying portfolio. Another problem is that the implied compound correlation might not be unique if we have a tranche with attachment and detachment around certain sections of the total portfolio loss distribution curve.

A more recent market development is to treat a CDO tranche as a derived security. The CDO tranche can be treated as a call or put spread on the total loss of the underlying portfolio. For example, long a 3-7% tranche protection on the credit index is equivalent to long a call option on the total loss with a strike price of 3% of the total notional amount of the underlying portfolio and short a call with a strike price of 7% of the total notional amount of the underlying portfolio. Then we just need to price all equity tranches at different detachment levels. Correspondingly the implied correlation could be given just for equity tranche with different tranche sizes or detachment levels. This way of quoting correlation and pricing CDOs are called the *base correlation method* introduced by J. P. Morgan. However, the base correlation method does not guarantee that it is an arbitrage free model. We have observed that for two tranches with the same tranche size, the one with more support or higher attachment level could have a break-even spread higher than the one with a lower attachment level in the base correlation framework. Also the base correlation method along with the Gaussian copula model does not tell us too much about how to price non-index portfolio tranches, neither more complicated structures such as CDO squared or CDO trades with both long and short credits as collaterals. We still need to use various loss mapping algorithms to derive a base correlation curve for the bespoke portfolio from the base correlation curve from an index tranche market. The reader is also referred to chapter 9 for a detailed discussion on the properties of the base correlation approach.

This chapter provides an overview about the use of the copula function framework in the context of credit portfolio modeling. We outline the essential steps when we apply the copula function. We provide some insights to default correlation from conditional perspective and study the impact of default correlation on the total credit portfolio loss distribution. We also provide some details with respect to CDO pricing when copula function is used.

The rest of the chapter is organized as follows: We introduce briefly in Section 2 our approach to characterize single name default swaps. In Section 3 we present our

copula function approach to default correlation, and also present different representation of the correlation structure in the Gaussian copula function. In Section 4 we examine the correlation parameters in a copula function from a conditional perspective to gain intuitive understanding of the correlation impact. We examine the conditional default swap spread if one or more credits default or survive to a specific time in the future. In Section 5 we examine the total loss distribution of a given credit portfolio and study its sensitivity to the correlation parameter. We also present two analytical computation methods for the total portfolio loss distribution. In the final section, Section 6 we explain in details how to use the model to price portfolio type transactions. The first example shows the pricing results of *i*th-to-default and the second one gives detailed formulas for CDO pricing. We conclude the chapter by certain remarks on the shortcoming of the copula function approach and possible extension in the future.

Default Information of a Single Name

To price any basket credit derivative structure, we first need to build a credit curve for each single credit in the portfolio, and then we need to have a default correlation model so that we can link all individual credits in the portfolio.

A credit curve for a company is a series of default probabilities to future dates. Traditionally we use rating agency's historical default experience to derive this information. From a relative value trading perspective, however, we rely more on market information from traded assets such as risky bond prices, asset swap spreads or nowadays directly the single name term structure of default swap spreads to derive market implied default probabilities. These probabilities are usually called ***risk neutral default probabilities***, which, in general, are much higher than the historical default probabilities for the rating class to which this company belongs. Mathematically we use the distribution function of survival time to describe these probabilities. If we denote τ as an individual credit's survival time which measures the length of time from today to the time of default, we use $F(t)$ as the distribution function defined as follows:

$$F(t) = \Pr[\tau \leq t] = 1 - S(t).$$

Where $S(t)$ is called the survival probability up to time t . The marginal probabilities of defaults such as the ones over one-year periods, or hazard rates in continuous term, are usually called a *credit curve*. In general, for single name default swap pricing only a credit curve is needed in the same way as an interest rate curve is needed to price an interest rate swap.

Correlated Defaults through Copula Function

Central to the valuation of the credit derivatives based on a credit portfolio is the default correlation. To put it in a simple term, the default correlation measures the impact of one credit default on other credits. Intuitively one would think of default correlation as being driven by some common macroeconomic factors. These factors tend to tie all industries into the common economic cycle, a sector specific effect or a company specific effect. From this angle it is generally believed that default

correlation is positive even between companies in different sectors. Within the same sector we would expect companies to have an even higher default correlation since they have more in common. For example, the recent overcapacity of telecommunication industry results in the default of numerous telecommunication and telephone companies. However the sheer lack of default data means those assumptions are difficult to verify with any degree of certainty. Then we have to resort to an economic model to solve this problem.

From a mathematical point of view, we know the marginal distribution of survival time of each credit in the portfolio and we need to find a joint survival time distribution function such that the marginal distributions are the same as the credit curves of individual credits. This problem cannot be solved uniquely. There exist a number of ways to construct a joint distribution with known marginals. Copula functions used in multivariate statistics provide a convenient way to specify a joint distribution with given marginal distributions.

A Copula function is simply a specification of how to use the univariate marginal distributions to form a multivariate distribution. For example if we have N correlated uniform random variables $U_1, U_2, \dots, U_N$, then

$$C(u_1, u_2, \dots, u_N) = \Pr\{U_1 < u_1, U_2 < u_2, \dots, U_N < u_N\}$$

is the joint distribution function which gives the probability that all of the uniforms are in the N dimension space cube. Using this joint distribution function C and N marginal distribution functions $F_i(t_i)$, which describes N credit curves, we form another function as follows $C[F_1(t_1), F_2(t_2), \dots, F_N(t_N)]$. It can be shown that this function could be a distribution function for the N random variable of survival times and its marginal distributions are $F_1(t_1), F_2(t_2), \dots, F_N(t_N)$, see Li (2000). So a copula function is nothing more than a joint distribution of uniform random variables from which we can build a joint distribution with a set of given marginals.

Then we need to solve two problems. First, which copula function should we use? Second, how do we calibrate parameter in a copula function? Suppose we study a credit portfolio of two credits over a given period. The marginal default probabilities are given by the two credit curves constructed using market information or historical information. From an economic perspective a company defaults when its asset falls below its liability. However, in the relative value trading environment we know the default probability from the credit curve constructed using market information such as default swap spreads, asset swap spreads or risky bond prices. Then we can assume that there exists a standardized asset return X and a critical value x , and when $X \leq x$ the company would default. i. e.

$$\begin{aligned}\Pr[X_1 \leq x_1] &= \Phi(x_1) = q_1 \\ \Pr[X_2 \leq x_2] &= \Phi(x_2) = q_2\end{aligned}$$

Where Φ is the cumulative univariate standard normal distribution. We use Φ_n to denote the n - dimension cumulative normal distribution function. If we assume that

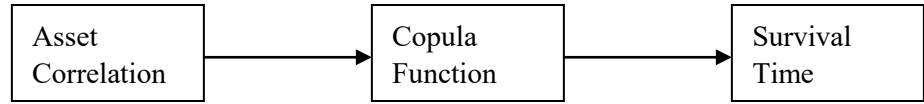
the asset returns follow a bivariate normal distribution $\Phi_2(x, y, \rho)$ with correlation coefficient ρ , the joint default probability are given by

$$\begin{aligned} \Pr[X_1 \leq x_1, X_2 \leq x_2] &= \Pr[X_1 \leq \Phi^{-1}(q_1), X_2 \leq \Phi^{-1}(q_2)] \\ &= \Phi_2[\Phi^{-1}(q_1), \Phi^{-1}(q_2), \rho] \end{aligned}$$

This expression shows that we can use normal copula function with asset return correlation as its parameters. Then we need an asset return correlation engine to get pair-wise correlation.

The above argument needs not to be associated with a normal copula. Any other copula function would be still able to give us a joint survival time distribution while preserving the individual credit curves. We have to use extra conditions in order to choose appropriate copula function. When we compare two copula functions we need to control the marginal distribution free correlation parameter such as the rank correlation.

This approach gives a very flexible framework based on which we can value many basket structures. It can be expressed in the following graph



We also present an efficient simulation algorithm here to implement our framework. To simulate correlated survival times we introduce another series of random

Variables $X_1, X_2, \dots, X_n$ such that

$$X_i = \Phi^{-1}(F(\tau_i))$$

Where $\Phi^{-1}(\cdot)$ is one-dimension standard normal inverse function. The set of random variables $X_1, X_2, \dots, X_n$ follow a joint normal distribution with a given correlation matrix Σ . From this equation we see that there is a one-to-one mapping between X_i and τ_i . Any problem associated with τ_i could be transformed into a problem associated with X_i , which follows a joint normal distribution. Then we could make use of efficient calculation method of multivariate normal distribution function.

The correlation parameters, in the framework of our credit portfolio model, can be roughly interpreted as the asset return correlation. However, in most practical use of the current model we either set the correlation matrix using one constant number or

two numbers as the inter and intra industrial correlation for trading models. We could either use an economic model to asset correlation or we can calibrate the parameters using traded instruments involving correlation such as first-to-default or CDO tranches.

The commonly used one or two correlation parameters are strongly associated with factor model in asset returns. For example, the one correlation parameter $\rho \geq 0$ corresponds to a one factor asset return model where each asset return can be expressed as follows:

$$X_i = \sqrt{\rho} \cdot X_m + \sqrt{1-\rho} \cdot \varepsilon_i$$

Where X_m represents the common factor return and ε_i is the idiosyncratic risk associated with credit asset i . Finger (1999) uses this one-factor copula for portfolio loss calculation. For a detailed discussion on this one-factor copula model, the reader is referred to chapter 8.

If we use two parameters, the inter industrial correlation ρ_o and the intra industry correlation ρ_I , then for each credit of industry group $k = 1, 2, \dots, K$ we can express the asset return as follows (see Li and Skarabot (2004))

$$X_i = \sqrt{\rho_I - \rho_o} \cdot X_k + \sqrt{\rho_o} \cdot X_m + \varepsilon_i$$

Using these factor models we can substantially reduce the dimensionality of the model. The number of independent factors then does not depend on the size of the portfolio. For example, for a portfolio whose credits belong to 10 industries we just need to use 11 independent factors, one factor for each industry and one common factor for all credits. We could substantially improve the efficiency of our simulation or analytical approach once we exploit the property of the factor models embedded in the correlation structure. Some other orthogonal transformations such as the ones obtained by applying principal axis component analysis could also be used to reduce the dimension.

Default Correlation – A Conditional Perspective

There are some concerns that the current model does not capture the "dynamic aspect" of spreads. It is believed that when one defaults, the spread of another one does not change since we use deterministic spread curves or credit curves. Here we use copula function to combine credit curves, and in fact this way of linking credit curves does capture one dynamic element of the spread, the part which is purely driven by the timing of default or survival.

One perspective to the correlation problem is to study the problem by **conditioning**, or in layman term "what happens to Brazil credit if Argentina defaults next month?". In fact, the ultimate objective of introducing correlation is to study the impact of one credit default on another credit default or spread. This section shows how to study the default correlation through conditioning. Specifically we give the conditional default probabilities or hazard rates of one credit when another correlated credit defaults or survives to a specific time in the future.

We use hazard rates to make comparisons of credit qualities. The hazard rate has a conditional probability density interpretation, which gives the value of the conditional default probability at a time interval of $t + \Delta t$, given survival to time t . Modeling a default process is equivalent to modeling a hazard function. There are a number of reasons why modeling the hazard rate function may be a good idea. For example, it provides us with the useful information of immediate default risk of each name known to be alive at any given time t . Moreover, the comparisons of different credits are most incisively made via the hazard rate function. For illustration purpose here we assume the time interval is one year, then the hazard rate can be interpreted as the probability of default over one year for small default probabilities.

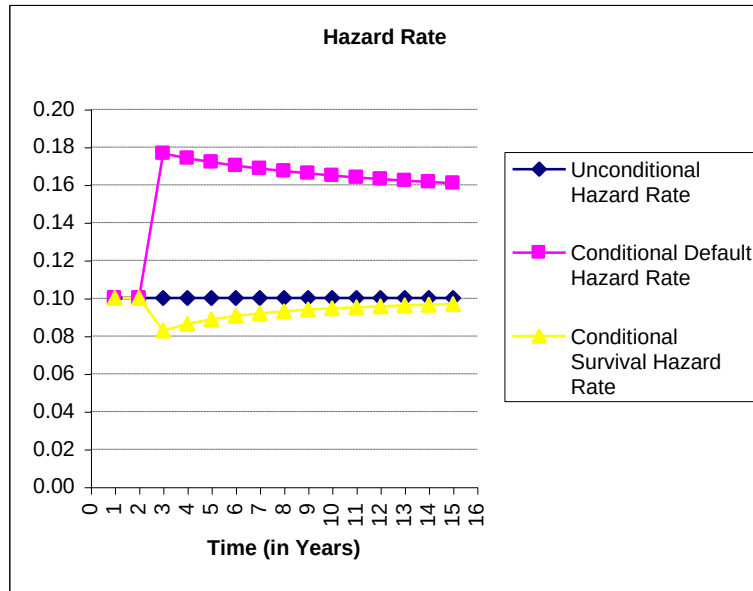
The conditional default probability or hazard would depend on the following few factors:

1. The conditioning part. A credit of high spread defaulting late has more information than a credit of low spread defaulting. Similarly a credit of high spread survives to a given point of time in the future has a higher impact than a credit of low spread. This shows the impact of timing of default or survival, and also the impact of relative credit quality.
2. Correlation parameter: In general the higher the correlation the higher the impact is

We show some numerical examples using the following assumption. The spread for credit 1 and 2 are 500 bps, recovery rate is 50%, which would result in about 10% default probability per year. The correlation parameter is 50%.

We plot in Figure 1 the conditional hazard rates of credit one under the following three cases:

Figure 1.

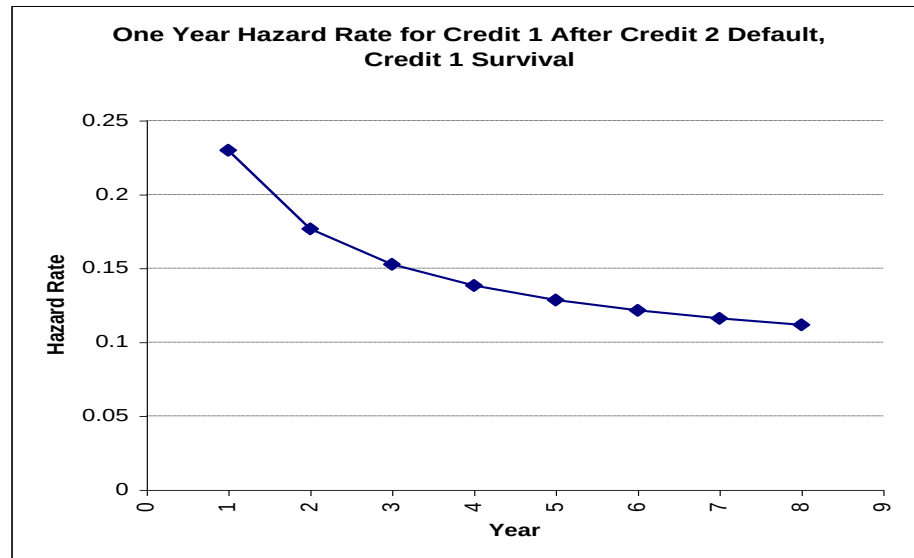


1. no conditioning at all, the hazard rate for credit one is given by the diamond line.
2. conditional on that credit two defaults at the end of second year and that credit one survives to that point, the resulting hazard rate of credit 1 is given by the square line
3. conditional on that both credits survives to the end of the second year, the resulting hazard rate of credit one is given by the triangle line

This example one shows that condition 2 has a larger impact than condition 3. This is not surprising since credit two has only about 20% probability of default over two year horizon, but it defaults now in condition 2 while the survival probability over two years is about 80%.

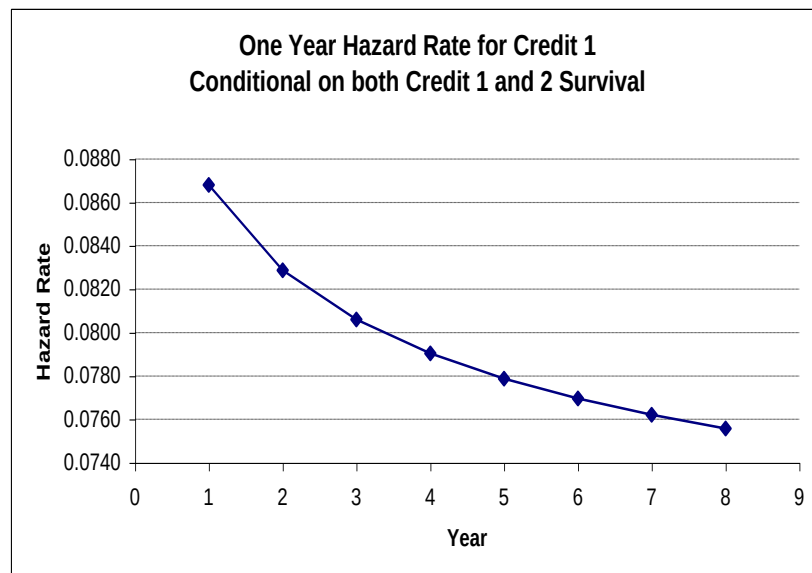
We also see that the impact on credit 1 hazard rate is decreasing as time passes. The reason is that the dynamics here is purely driven by “survival bias”, which we assume deterministic. This shock will be taken over by its own credit quality as time passes.

Figure 3 gives the hazard rate of credit one when credit two defaults at different times in the future and credit 1 survives to that time.

Figure 2.

This graph shows the impact of timing of conditioning. We see that the one-year hazard rate of credit one immediately after credit two defaults decreases as the credit two defaults further in the future. Credit two has about 10% default probability. On expectation credit two would have defaulted by the end of the 10th year. So if we assume it defaults at year 10 it should have basically no impact at all on credit 1. But if we assume credit 2 defaults in year 1 when it has only 10% default probability over the period, it would have a big impact on credit 1.

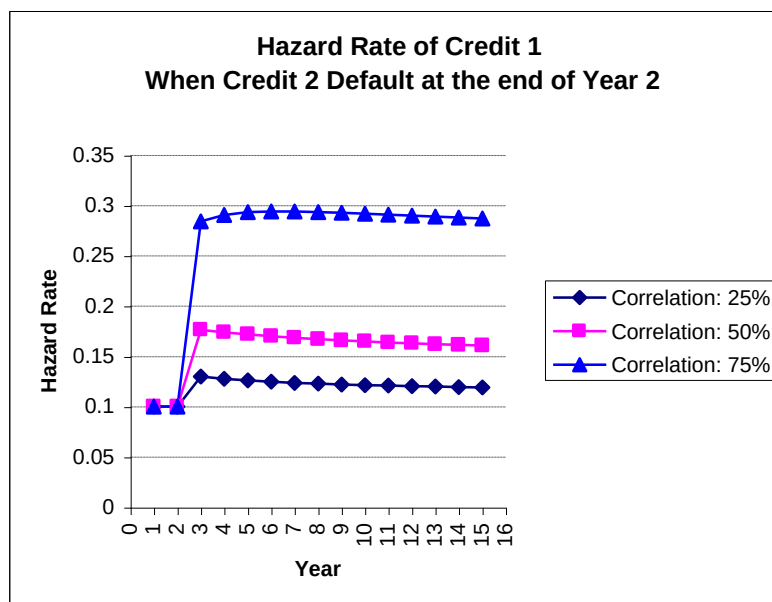
Figure 3 shows one-year hazard rates of credit one when both credits survives to a given time in the future.

Figure 3.

This graph shows that credit one's default probability will decrease if both credits survives to a given point time in the future. It has a higher impact if neither defaults over a longer period of time than over a shorter period of time.

Figure 4 shows the correlation impact of credit one's hazard rate when credit 2 defaults

Figure 4.

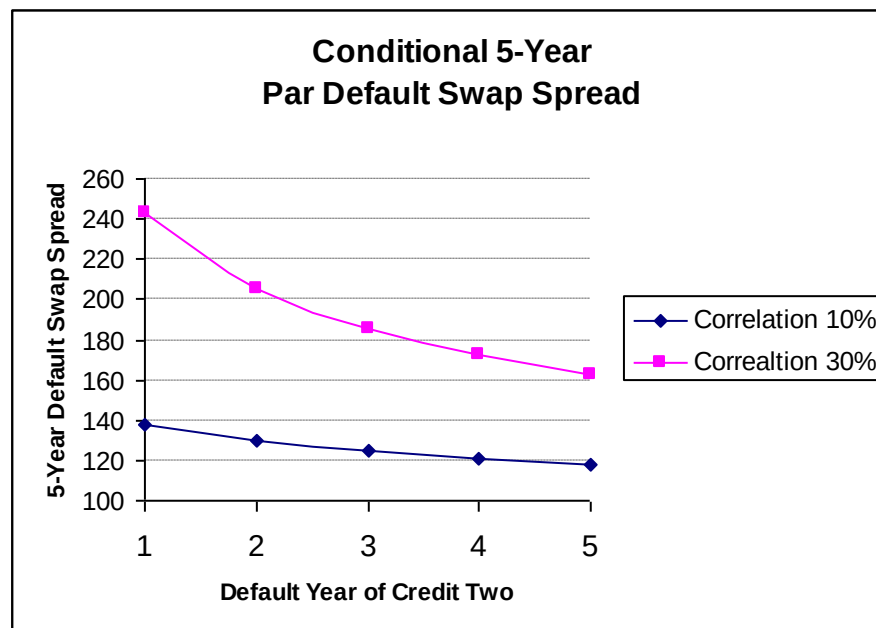


This graph corresponds to Figure 1's square line with different correlation assumptions. It shows that one year default probability of credit 1 increase by 30% immediately after credit two defaults when correlation is 25%, by 76% and 184% when correlation is 50% and 75%. This shows the non-linear impact of correlation on the default probability of credit one when credit two defaults at a specific time in the future. The process of decay will be much longer when correlation is high than when correlation is low. This also shows the importance of correlation when we model credit portfolio.

All these properties of conditioning can be translated directly into par default swap spreads. The following graph shows the conditional 5-year default swap spread on credit one when credit defaults or survives to various points of time in the future.

Figure 5 shows the 5-year conditional par default swap spread on credit one when credit two defaults at different points of time in the future. The original 5-year default swap spreads for both credit one and two are 100 bps. After credit two defaults, the conditional 5-year default swap spread for credit one goes up. The closer that credit two defaults, the higher the conditional par swap rates of credit one goes up. The graph also shows the impact of correlation parameter. The conditional 5-year par swap rate for credit one would go from 100 bps to 140 bps when the

correlation parameter is 10% and credit two defaults at the end of the first year. The same number would go up to 240 bps if the correlation parameter is 30%. This gives us an intuitive way of calibrating correlation parameter in the normal copula function. In practice we have some general idea about how much the par default swap spread of one credit could change if another credit defaults. Using this intuitive conditional par default swap spread we can solve for the correlation parameter in the copula function. In the normal copula framework we can study the correlation parameter pair by pair.



The Property of Loss Distribution

For a given credit portfolio the first question investors would like to know is its loss distribution over a given time horizon in the future such as 1 year or 5 years. This would give the investor some ideas about the possible default loss of his investment in the next few years. The information we need to use in our framework is as follows: the credit curve of each credit that characterizes the default property over the time horizon, the recovery assumption and the asset correlation structures. Many useful risk measurements, such as the expected loss, the unexpected loss or the standard deviation of loss, the maximum loss, VaR or the conditional shortfall, could be obtained easily once the total loss distribution is calculated.

Here we study the property of the loss distribution using a numerical example. The base case used is as follows:

Figure 5.

Number of Asset	100
Credit Spread	200 bps
Correlation	50%
Maturity	5 year
Recovery Rate	30%

Source: Salomon Smith Barney.

Figure 6 shows the excess loss distribution where the x-axis is the loss amount and y-axis is the probability of loss more than a given amount in the x-axis. All excess loss functions would start from 1.0 and gradually go to zero. If we include the zero loss in the probability calculation, then the probability of having non-negative losses is always 1. We purposely exclude the zero loss in the calculation so that we can see the probability of having zero loss in the graph explicitly. Let's define the excess loss more precisely. Suppose that L represents the total loss of the portfolio which is a random variable since we do not know for sure what value it takes. For a given set of loss amount $l_0, l_1, \dots, l_n$, we can calculate the probability of excess loss $p_0, p_1, \dots, p_n$ as follows:

$$p_i = S(l_i) = \Pr[L > l_i]$$

The excess loss distribution essentially depicts (l_i, p_i) . The reason we use excess loss distribution instead of loss distribution, which is defined as $F(l_i) = 1 - S(l_i)$, is mainly due to the fact that many interesting properties of the loss distribution can be viewed more explicitly from the excess loss distribution graph than from the ordinary loss distribution graph. For example, the expected loss using the density function $f(l)$ of the loss distribution can be calculated as follows:

$$\mu_L = E(L) = \int_0^{\infty} l \cdot f(l) dl = \int_0^{\infty} S(l) dl,$$

which is just the area below the excess loss distribution line. Some other quantity such as the expected loss of tranch securities (loss with a deductible and a ceiling) could also be more simply expressed if we use excess loss function. We will get back to this point in the next section when we discuss about CDO pricing.

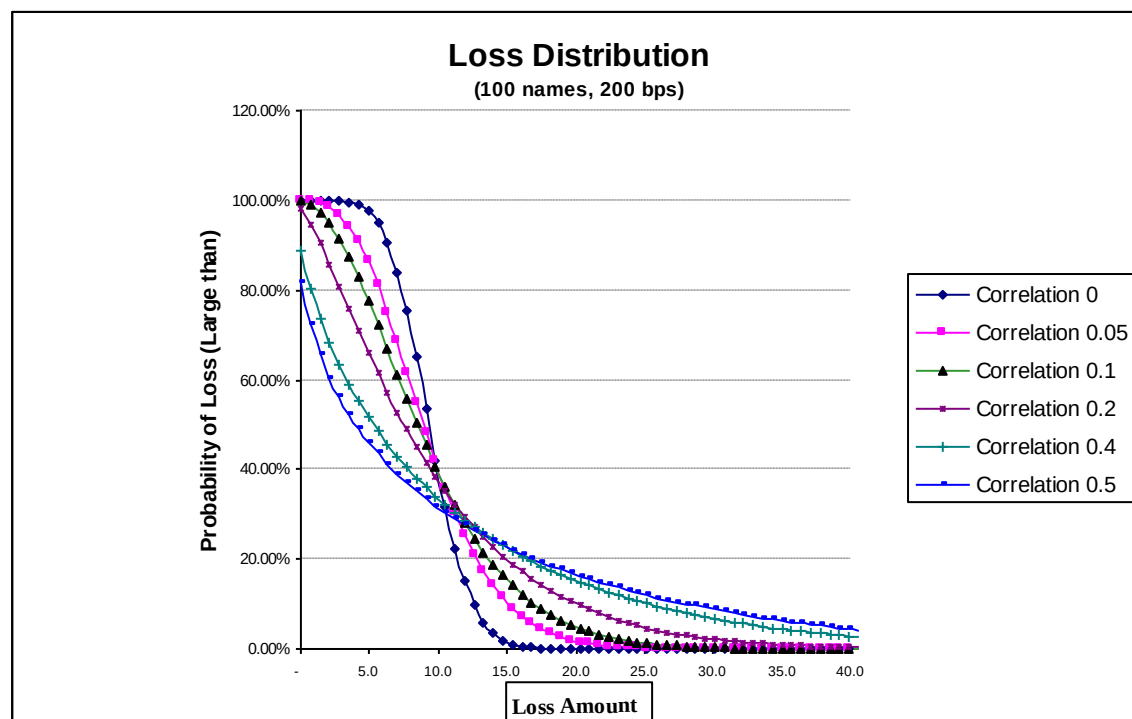
Figure 6.

Figure 6 shows the impact of correlation on the total excess loss distribution. From the graph we see that the probability of having zero loss goes from almost 0 to about 20% when correlation changes from 0 to 50%. The default probability over 5 years for each name is

$1 - e^{-5 \cdot 0.2 / (1 - 30\%)} = 13.31\%$, the probability of having no default of a portfolio with 100 independent names is $(1 - 13.31\%)^{100}$, a practically 0. However when the correlation is high, default occurs more in a bulk, which makes the probability of having zero loss go up to 20%. When correlation is high, more loss would be pushed to the right, which makes the excess loss distribution tail much fatter since the expected total loss, the area below the excess loss function line, does not change along with the change of the correlation. This can be shown using a credit VaR, which is defined as the excess loss that probability of loss larger than this value is less than a given percentage such as 1%. The 1% credit VaR for various correlation values are given in the following table.

Figure 7.

Correlation	C-VaR
0%	14.7
10%	25.2
20%	32.9
50%	53.9
75%	67.2

Source: Salomon Smith Barney.

In practice how to quickly obtain an accurate total excess loss distribution is very important. There are a variety of methods that have been used for the total loss distribution. Here we present the details on the recursive method in a one factor Gaussian Copula model and briefly summarize the conditional normal approximation approach.

We consider a credit portfolio consisting of n underlying credits whose notional amounts are N_i and fixed recovery rates are R_i , $i = 1, 2, \dots, n$. We consider the aggregate loss from today to time t as a fixed sum of random variables X_i :

$$L_n(t) = \sum_{i=1}^n l_i(t) = \sum_{i=1}^n (1 - R_i) \cdot N_i \cdot I_{(\tau_i \leq t)}$$

Where τ_i is the survival time for the i th credit in the credit portfolio and I is the indicator function, which is 1 in case of $\tau_i \leq t$ and 0 otherwise. The distribution function of survival time c is denoted as $F_i(t) = \Pr[\tau_i \leq t]$. The specification of the survival time distribution $F_i(t)$ is usually called a credit curve, which can be derived from market credit default swap spreads.

From the above equation we can calculate the total loss distribution as

$$\begin{aligned} F_{L(t)}(x) &= \Pr[L_n(t) \leq x] = \Pr\left[\sum_{i=1}^n X_i \leq x\right] \\ &= \int \Pr\left[\sum_{i=1}^n X_i \leq x \mid F\right] \cdot dF \end{aligned}$$

Conditional on common factor F , all X_i are independent, then we just need to calculate the convolution of n independent random variables, c . Based on the last section we know that X_i are independent conditional on the common factor X_M in the one factor model. Each X_i can take only two discrete values with constant recovery rate assumption as follows: the loss would be zero if default does not occur or $B_i = (1 - R_i)N_i$ if default occurs.

$$f(x \mid F) = \begin{cases} 0, & 1 - q_i(t \mid F) \\ B_i, & q_i(t \mid F) \end{cases}$$

where $q_i(t \mid F)$ is the conditional default probability for credit i before time t .

The density of the conditional total loss distribution can be calculated recursively over the partial sum $L_j = L_{j-1} + X_j$. We then have the following recursive formula

$$f_{L_j}(x \mid F) = \begin{cases} p_j \cdot f_{L_{j-1}}(x \mid F), & x < B_j \\ p_j \cdot f_{L_{j-1}}(x \mid F) + q_j \cdot f_{L_{j-1}}(x - B_j \mid F), & x \geq B_j \end{cases}$$

This has been described in Klugman, Panjer and Willmot (1998) and also in Anderson and Sidenus (2003). The unconditional total loss distribution is obtained by simply integrating the conditional loss distribution over the common factor F . In the simple case of one factor Gaussian copula model, we use a Gaussian quadrature for the integration over the one common factor. In the one parameter case, the conditional default probability can be calculated directly as follows:

$$\begin{aligned} q_i(t | X_M) &= \Pr[\tau_i < t | X_M] \\ &= \Pr[F_i^{-1}(N(X_i)) < t | X_M] \\ &= \Pr[X_i < N^{-1}(F(t)) | X_M] \\ &= N\left(\frac{N^{-1}(q_i(t)) - \sqrt{\rho} \cdot X_M}{\sqrt{1 - \rho}}\right) \end{aligned}$$

where $q_i(t)$ is the unconditional default probability of credit i before time t .

Another approach uses the conditional normal approximation. Conditional on the common factors all credits are independent. Based on the law of large number the total conditional loss distribution can be approximated by a normal distribution. This mean and variance of this normal distribution can be simply calculated similarly as we do in the above one factor case. More details are given as follows.

Conditioning on the common factor X_M we can compute the mean and variance of the total loss variable, $L | X_M$,

$$\begin{aligned} M_v &= \sum_{i=1}^n N_i \cdot (1 - R_i) \cdot q_i(t | X_M) \\ \sigma_v^2 &= \sum_{i=1}^n N_i^2 \cdot (1 - R_i)^2 \cdot q_i(t | X_M)(1 - q_i(t | X_M)) \end{aligned}$$

The conditional Normal approach uses Normal distributions to approximate the conditional loss distribution. The Normal distribution has the same mean and variance as computed above. In general, other distributions, such as inverse normal or Student-t, can be used. The normal distribution is chosen due to the central limit theorem, which states that the sum of independent distributions (but not identical distribution) approaches to a normal distribution as the number of the independent distributions increases. In this case the independent distributions are the distributions of the indicator functions of $N_i \cdot (1 - R_i) \cdot I_{[\tau_i < t]}$, which are independent when conditioned on the common factor X_M .

Given the conditional normal approach the conditional expected loss for a tranche with attachment and detachment K_L^T and K_U^T can be easily computed in closed form as follow:

$$E(L^T(t) | X_M) = (M_v - K_L^T) \Phi\left(\frac{M_v - K_L^T}{\sigma_v}\right) + \sigma_v \cdot \phi\left(\frac{M_v - K_L^T}{\sigma_v}\right) - (M_v - K_U^T) \Phi\left(\frac{M_v - K_U^T}{\sigma_v}\right) - \sigma_v \cdot \phi\left(\frac{M_v - K_U^T}{\sigma_v}\right),$$

where ϕ is the one-dimensional normal density function.

With the calculated conditional expected loss the unconditional expected loss is obtained simply by integrating over the common factor X_M .

$$E(L^T(t)) = \int_{-\infty}^{+\infty} E(L^T(t) | y) \cdot \phi(y) dy.$$

However by choosing normal distribution in its approximation the approach has its limitations. First of all a normal variable can have a negative value with no-zero probability. As we know the loss in a portfolio should never be negative. However this limitation only affects the equity tranche (the most junior tranche) and can be mitigated through method described below. Second as a loss is a summation of discrete loss variables, when a portfolio consists of only a few underlying names then approximating the loss by a continuous variable (such as a Normal variable) might not be a good approximation. This limitation also apply to some extreme case when the loss is dominated by only a few underlying names. In general the conditional normal approach is a very good approximation when the number of names in a portfolio is larger than 30. Most of CDO portfolios have number of names larger than 30.

To mitigate the negative loss problem for an equity tranche one can use the following method, which preserves the expected loss of a CDO portfolio. An equity tranche $[0, K_U^T]$ with detachment point K_U^T has payoff:

$$L^T(t) = L(t) - \max[L(t) - K_U^T, 0]$$

The conditional expected loss for the equity tranche is,%

$$E(L^T(t) | X_M) = M_v - (M_v - K_U^T) \cdot \Phi\left(\frac{M_v - K_U^T}{\sigma_v}\right) - \sigma_v \cdot \phi\left(\frac{M_v - K_U^T}{\sigma_v}\right)$$

This has been proven to work well for index equity tranche of size more than 2%. Alternatively we can also use inverse Gaussian distribution to approximate for the equity tranche since inverse Gaussian distribution takes only positive value.

Basket Credit Derivative Pricing

We present two examples to show how to price credit derivative using our copula function framework. The first example is an i th-to-default contract and the second one is on CDO transaction.

1. The Valuation of the i th-to-default Contract

The i th-to-default contract depends on the order of survival times of names in the basket. The premium leg pays a fixed premium as long as the i th-to-default does not occur. So the premium leg purely depends on the order statistics of survival times of the basket credits. However, the loss leg also depends on the identity of the i th-to-default credit if each name in the basket have different notional amounts or different recovery rate. Only in the case that all names have the same recovery rates and notional amount can we use simply the order statistics for both premium and loss leg calculation. Here we introduce a recurrence equation for the calculation of the order statistics for non-identical and non-independent variables as follows (Sathe and Dixit (1990)):

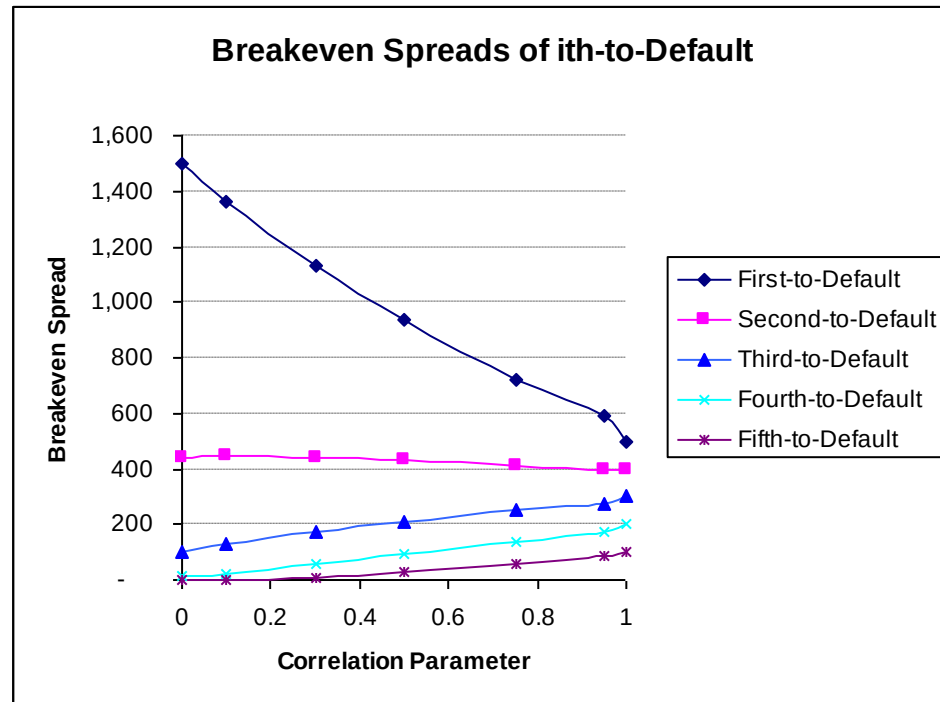
$$r \cdot F_{r+1,n}(x) + (n-r) \cdot F_{r,n}(x) = \sum_{i=1}^n F_{r,n-1}^{(i)}(x), 1 \leq r \leq n-1.$$

Where $F_{r,n}(x)$ is the distribution of r th order statistics $X_{r,n-1}$ out of n variables $X_1, X_2, \dots, X_n$, and $F_{r,n-1}^{(i)}$ is the distribution function of $X_{r,n-1}$ in a sample of sizes $n-1$ obtained on dropping X_i from the original sample of size n . From this equation we see that we can calculate the second-to-default, third-to-default etc once we have the first-to-default of the original basket minus one credit. This can be easily calculated once we condition on certain common factor, all names are independent from each other. Then we can easily calculate the first-to-default of $n-1$ names, and use the above formula to calculate the second-to-default, or any higher order of default in a recursive way. The loss leg can be similar calculated since in the independent case we can calculate the order statistics along with the identity. Details are omitted here due to space limit.

We produce a numerical example to illustrate how our method can be used to value a specific contract. We assume that the contract is an i th-to-default contract of a small basket of m credits. The number of credits here ranges most likely from two names to about 8 names. In this contract, the protection buyers pays a fixed premium conditional on the non-occurrence of the i th-to-default, in return, he would receive payment if the i th-to-default occurs. The payment is associated with the defaulted identity of m credit basket. For simplicity we assume that interest rate is constant 5% and credit spreads ranges from 100 bps to 500 bps for a credit of 5 names. The contract is 5-year with premium paid semiannually. The following table and graph shows the breakeven spreads for the i th-to-default contract.

Correlation	First-to-Default	Second-to-Default	Third-to-Default	Fourth-to-Default	Fifth-to-Default	Total Spreads
0%	1,500	442	102	13	1	2,058
10%	1,362	445	129	25	2	1,964
30%	1,131	442	174	55	10	1,813
50%	938	432	211	90	26	1,697
75%	723	412	250	140	59	1,584

95%	593	397	275	176	87	1,528
100%	500	400	300	200	100	1,500



First, let us examine the boundary condition for correlation parameter. When all 5 names are independent, the first-to-default breakeven spread is the sum of the all underlying credit spreads, which is 1500 ($=100+200+300+400+500$) bps. When all 5 names are perfectly correlated, the first-to-default(FTD) would pay the highest spread, 500 bps, the second-to-default would pay the second highest spread, 400 bps, ..., and the last-to-default would pay the lowest spread, 100bps. The impact of correlation parameters on different *i*th-to-default contracts is quite different. The FTD breakeven spread is strongly negatively related to the correlation parameter, the second-to-default breakeven spread is almost invariant to the correlation parameter changes and the third-to-default, fourth-to-default and the fifth-to-default are positively related to the correlation parameter. Another observation is that the total spreads paid on the *i*th-to-default contracts is larger than the total spread of all underlying credit spreads. This might seem surprising since buying all *i*th-to-default protection and buying all single name default protection would provide the same coverage from the loss protection perspective. But each of the contracts has two legs: the premium payment leg and the loss protection leg. Different premium payments are based on different survival curve. For example even though FTD breakeven spread is high, but on expectation the default probability of FTD is also high, which makes the potential premium payment shorter than the maturity. Since the total expected discounted loss on the single name side is equal to the total expected loss on all *i*th-to-default contracts, the total premium paid on all single names would also

be equal to the total premium paid on the i th-to-default contracts. From the graph we can also see that FTD is a highly leverage position.

Second, we examine the relative value of the first-to-default, second-to-default and third-to-default contracts. Obviously as we go up the order of the default the contract should have a smaller value. But as the number of the underlying credits in the portfolio increases the difference between these contract values should also decrease.

2. The Pricing of a CDO Transaction

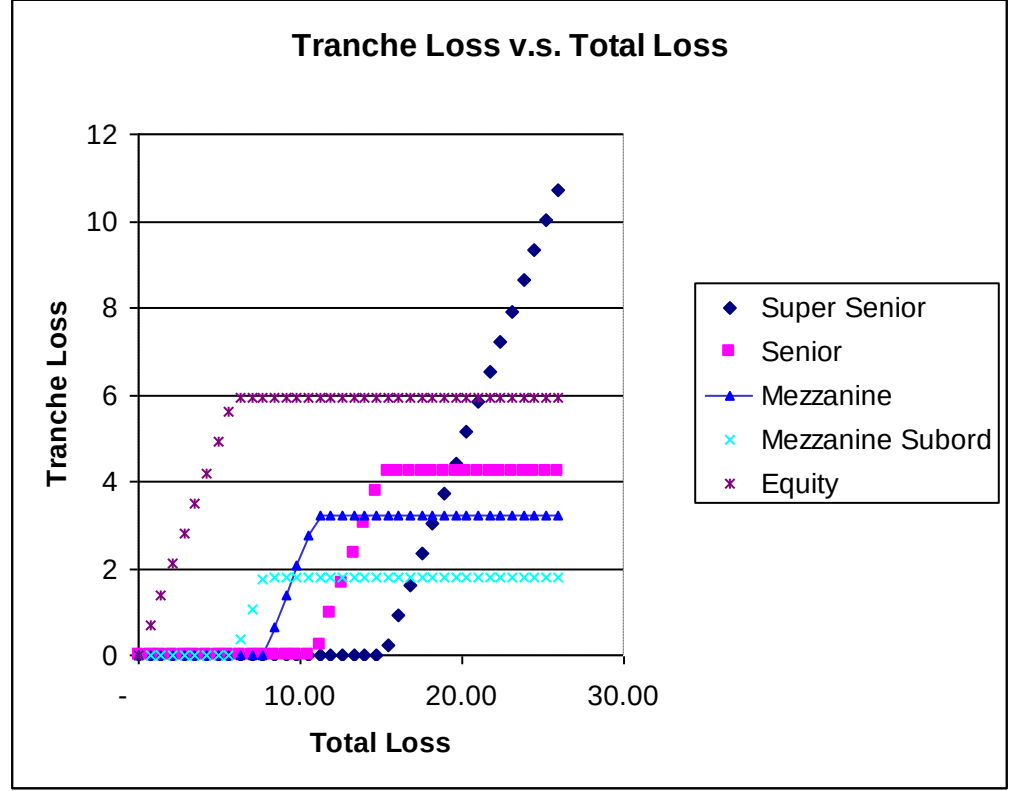
CDOs come in different shapes and forms. To illustrate our methodology, we use a simple synthetic CDO structures in which the investor of a specific tranche i provides default protection against the total loss (L) of a portfolio over a certain range $[B_i, U_i]$. The payoff function of the protection is as follows:

$$P_i = \begin{cases} 0, & L \leq B_i, \\ L - B_i, & B_i < L \leq U_i, \\ U_i - B_i, & L > U_i. \end{cases}$$

This is similar to an insurance contract with deductible of B_i and ceiling of U_i . The premium side is assumed to be paid at a flat rate applied to the outstanding principal of the tranche.

We can look at this payoff function from an option perspective. The following graph depicts the payoff function of a 5-tranche CDO against the total loss distribution. Buying a super senior tranche is equivalent to selling a call option on the total loss of the portfolio with a strike of the lower subordination level of the super senior tranche. Buying an equity tranche is equivalent to selling a put option with the strike price equal to the equity size. The investor of any middle tranche is equivalent to having an option strategy of *spread* in which he is long a put option with strike price of U_i and short a call with strike price of B_i . Looking at a tranche from this angle gives us alternative views to the properties of CDO tranches.

Figure 8.



From valuation perspective we just need to value both premium leg and loss protection leg. We assume that the accumulative tranche loss up to time t is $L^T(t)$, with the attachment level K_L^T and detachment level K_U^T , and the tranche size is $\Delta^T = K_U - K_L$. If we ignore the interest rate impact the payoff and its expected value can be expressed as follows:

$$L^T(t) = \max(L(t) - K_L, 0) - \max(L(t) - K_U, 0)$$

$$E(L^T(t)) = \int_{K_L}^{K_U} S_L(x) dx,$$

The expected tranche loss is the area of the excess loss distribution bounded by the subordination levels K_L^T and K_U^T if we ignore the interest rate. Once we consider the impact of the interest rate we have to resort to some numerical method for the calculation, and we do the calculation over each premium payment period such as 3-month period.

The premium and loss sides depend on the evolution of outstanding principal of the CDO tranches, which is easy to calculate since, in our recursive or conditional normal approximation approach, we know the cumulative loss distribution up to any time in the future. When we use a base correlation approach we simply use the correlation corresponding to each detachment point for the loss distribution

calculation, and then take the tranche loss as the difference of the two equity tranches loss calculated with different correlation numbers.

Synthetic CDOs tranches is a portfolio transaction between two parties, the protection buyer and the protection seller. The protection seller gets paid a fixed percentage of the outstanding notional amount of the chosen tranche, and incurs a loss only after the accumulative loss of the total portfolio becomes more than the attachment point, but the loss of the tranche investors is capped by the size of the chosen tranche.

We first obtain the unconditional tranche expected loss $E(L^T(t))$ across different time horizon t , then we can calculate both default leg and premium leg of a CDO tranche. Let T_1 and T_2 be the beginning and the end of a premium payment period, then the present value of the tranche expected loss for this period is

Here $\bar{O}_t = \Delta^T - E(L^T(t))$ is the tranche outstanding notional at time t . Assuming a log-linear dependency between $\bar{O}_{t_1}$ and $\bar{O}_{t_2}$ over the time interval $[T_1, T_2]$, we can write

$$\begin{aligned}\bar{O}_{T_2} &= \bar{O}_{T_1} \cdot \exp(-h(t - T_1)) \\ h &= \frac{\log(\bar{O}_{T_1} / \bar{O}_{T_2})}{T_2 - T_1}\end{aligned}$$

Here h can be viewed as the forward tranche "hazard rate" for the loss over the time interval $[T_1, T_2]$. Then the default leg present value for this period is,

$$DL[T_1, T_2] = \frac{h}{h + f} [df(T_1)\bar{O}_{T_1} - df(T_2)\bar{O}_{T_2}]$$

Where $df(T)$ is the risk-free discount factor at time t , $f = \frac{\log(df(T_1)/df(T_2))}{T_2 - T_1}$

is the forward risk free rate over the period.

The premium leg for this period with market standard continuous settlement (accrual coupon is paid when default occurs) is,

$$\begin{aligned}PL[T_1, T_2] &= \pi^T \cdot \Delta T \cdot df(T_2) \cdot \bar{O}_{T_2} + \pi^T \cdot \frac{h}{h + f} \cdot \\ &\quad \left\{ \frac{1}{h + f} [df(T_1)\bar{O}_{T_1} - df(T_2)\bar{O}_{T_2}] - \Delta T \cdot df(T_2) \cdot \bar{O}_{T_2} \right\}\end{aligned}$$

Where π^T is the premium coupon paid over the period $[T_1, T_2]$.

This calculation is done over each period. Then, we can calculate the present values of the premium leg and loss leg by simply adding the values for all periods.

Both tranche default leg value and premium leg value are completely determined by the tranche expected loss (or equivalently tranche outstanding notional) along the time horizon. We can define tranche survival probability as the relative outstanding

notional (outstanding notional divided by the original tranche size) then the tranche pricing is the same as a single name default swap pricing assuming zero recovery. All techniques in the single name CDS pricing can be used. As all the derivations in this section do not use any correlation assumption, one may also directly describe the portfolio loss process and use formulas here to compute CDO prices.

We give a numerical example of our CDO pricing model. We assume that there are 100 credits with an average 5-year spread of 80 bps. The subordination levels are as given in the following table. The correlation parameter used is a pairwise constant correlation of 12.50%.

Name	Notional %	Expected Loss	Breakeven Spread
Super Senior	88%	281325.054	1
Senior	4%	1425416.793	77
Senior Mazzanine	3%	4306067.995	323
Junior Mazzanine	2%	6987660.768	877
First Loss	3%	22344375.23	3104

The above table also shows the breakeven spreads for all tranches as well as the expected loss for each tranche. While these numbers are perfectly right from a modeling perspective based on the model assumptions we make, but it does not produce breakeven spreads currently traded in the market. There are a variety of reasons for the discrepancies. First, we assume that the total spread for each name is purely due to default. In reality we understand that is a large proportion of credit spreads are attributable to non-default factors such as liquidity etc. Second, most of market participants are using historical information in their pricing of credit portfolio transaction by adding certain risk premium, different participants might have used totally different risk premiums. Third, the copula function we used is just one of the many possible transformation functions we have from single name credits to portfolio credits. Some alternative copula functions might produce better fit to what is traded in the market, see Li and Liang (2005) for an extension to use a mixture Gaussian copula model. Last, for the moment we do not incorporate stochastic spreads into our framework, this could have some potential impact on the final pricing results. But this does not prevent us from using the Gaussian copula along with a base correlation curve to mark to market our daily P&L for CDO transaction. This is similar to that we use a Black-Scholes formula with a volatility curve or surface instead of a constant volatility to price an option book.

Basket Credit Derivative Pricing: From Index Tranche Pricing to Bespoke Portfolio Tranche Pricing

The current market benchmark method is to use Gaussian copula function along with a base correlation method. In this framework, there is a one to one relationship between the tranche spread and implied base correlation curve. From a given set of index tranche spreads we can use calibration to obtain a base correlation curve. However it does not tell us how to price bespoke portfolio tranches using a base correlation curve obtained from the index tranche market. Practitioners in the market employ various mapping algorithms based on the bespoke portfolio loss characteristics relative the index portfolio loss to derive a bespoke portfolio base correlation curve from the index tranche market's base correlation curve. We provide a summary here and some general comments.

In the original copula function framework we intend to study a portfolio problem in two stages. First, we use a credit curve to describe a single name default property. Second, we use a copula function to describe the default correlation problem. The correlation structures and correlation parameters are independent from the magnitude of single name default probabilities. That is why we say that the correlation parameter is more rank type correlation which provides an order of default. In practice we often use a single correlation parameter. We have to emphasize that it is not the original framework that allows a single correlation parameter, but simple a convenient choice made by practitioners so that it is easy to communicate with each other about the correlation.

Under these constraints it is pretty hard to compare one credit portfolio against another credit portfolio with respect to the correlation. We simply try to link the correlation with the total loss distribution characteristics. Here are three most commonly used base correlation mapping methods.

1. Normalized Strike (NS).

In this approach we simply link the correlation based on the expected loss of the portfolio. The intuitive idea is as follows: a mezzanine tranche of a bespoke credit portfolio with higher spreads should behave like an equity tranche of the index. Then we can use the following mapping approach

$$\rho_{bespoke}(K) = \rho_{index} \left[\frac{EL_{index}}{EL_{bespoke}} \cdot K \right]$$

This is easy to implement. However, it does not fully consider the dispersion of a credit portfolio relative to the index. A few credits with a higher spreads could distort the scaling substantially. Also, comparing two portfolios in terms of their expected values is a very simplistic comparison.

2. Normalized Loss Ratio

This approach uses the equity tranche loss, instead of the portfolio expected loss, for the base correlation mapping. The formula is as follows

$$\frac{EL_{bespoke}[0, K] \text{ with } \rho_{bespoke}(K)}{EL_{bespoke}} = \frac{EL_{index}[0, K] \text{ with } \rho_{index}(K)}{EL_{index}}$$

This is still easy to implement. It makes the use of equity tranches loss with different sizes in addition to the portfolio expected loss. From the comparison of two portfolios in terms of loss distribution perspective, it provides more detailed comparison than the normalized strike method, but still does not capture the loss variations within a tranche. Sometimes it is impossible to find a solution for this method.

3. The market simply quotes the implied based correlation.

$$\Pr(L_{bespoke} > K_{bespoke}; \rho_{index}(K)) = \Pr(L_{index} > K; \rho_{index}(K))$$

This approach compares the loss distribution of bespoke portfolio against that of an index portfolio directly by the percentile-to-percentile comparison. We simply find a corresponding strike for the bespoke portfolio whose excess loss probability at the strike (to be solved) is equal to that of the index portfolio while both excess probabilities are calculated at the same index base correlation at the index strike level. From loss distribution comparison perspective, it provides the most comprehensive comparison. Both average spread and the dispersion of spread should be captured by this method.

While all these mapping algorithms based on loss distribution are intuitive, none of them is fully theoretically justifiable. Without fully use of individual name information such as asset return volatilities and correlations it is hard to compare one portfolio against another portfolio as we do in equity portfolio.

Risk Measurement and Hedging

Once a model and a mapping algorithm are chosen we can price all credit portfolio trades, and produce a series of risk measures based on the model. These risk measures are then used to form a hedging strategy for the trading book. The common used risk measures are as follows:

- Credit spread delta is defined as the sensitivity of a trade to the instantaneous movement of the spread of a single entity, with all other parameters remaining constant. It is calculated through perturbation of the individual credit curves. Individual spread delta is reported as the change in value of the trade for a 1 basis point (1 bp) move in the indicated spread. Individual spread delta can be calculated as parallel moves in the individual curve (in which each spread on a particular curve is moved by 1 bp in a parallel fashion) or as bucketed moves in the curve. Individual spread delta is calculated trade-by-trade, and aggregated based on issuer name, industry or portfolio level. When we change the spread we recalibrate the credit curve or instantaneous marginal default probability. Global spread delta is defined as the change in the portfolio value change when all the underlying reference credit curves move by 1 bp. Global spread is calculated by bumping all

spread curves of the underlying reference credits simultaneously in a parallel way or in bucket. Sometime we also study the sensitivities of our trade or book with respect to large spread movement. Another common practice is to adjust the individual spread movement with respect to the index spreadsheet. The reason is that not all individual spread moves by the same amount when index moves. A statistical beta based on regression analysis is usually used.

- Individual spread Gamma is defined as the sensitivity of individual spread delta to a 1 bp move in a particular reference credit. As such it represents the second-order price sensitivity with respect to a change in the spreads of the reference credit. Individual spread Gamma is calculated by bumping one credit curve a time while all other credit curves remain the same for portfolio transactions. Global gamma is defined as the change in the global spread delta (which is defined as the portfolio value change when all the underlying reference credit curves move by 1 bp) of a portfolio for a 1bp move in all reference credit spreads simultaneously. Global spread gamma is calculated by bumping all spread curves of the underlying reference credits simultaneously in a parallel way. Sometime we simply use a large spread movement as a measure of gamma risk by bumping the current spread by 50%. Similar to spread delta risk we can also use bucket gamma risks which are more computationally challenging.
- Jump-to-default risk: We measure it by simply assuming one name defaults right away or at specific time in the future. We can also study the group jump-to-default risk.
- Time-decay risk. This measure the risk that as time passes, or maturity shortens, the value of portfolio transactions changes. For portfolio credit default swap its survival time curve is most likely not flat which makes the time decay an important risk factor.
- Correlation risk: Since we use a base correlation curve, we could measure the risk in terms of parallel change or bucket correlation change. In practice we very often see a correlation curve twist which reflects market changing perception about different tranche risks. We can measure this risk by creating a sensitivity report for the whole book with respect to each base correlation point.

In practice we tend to minimize spread and gamma risks, control jump to default risk and also make correlation risk flat. We would also like to have positive carry: we receive more cash inflow than outflow. The hedging instruments we use are single name and index credit default swaps and index tranches. Very often broker dealer tends to run some residual risks by hedging more than model required amount to maintain positive carry. But this strategy does not work well all the time, especially during turmoil when there are out-of-the-blue defaults or large and jumpy spread movement. We can use index tranche to hedge the base correlation risk. Sometime we can also use the index plus complementary tranches to hedge the correlation risk. In conclusion, for any hedging strategy there are still residual risks left. What

residual risks a desk should keep is the key which depends on the trader's judgment of the market and trader's own experience.

Conclusion

In conclusion the Gaussian copula function approach along with a base correlation method provides a simple and flexible framework to price basket credit derivatives. We studied further the framework and gained some more insights of it, especially from the conditional perspective of its correlation structure. This shows that the Gaussian copula function implies a too strong correlation structure. The reason for this is that we describe each credit using only two states: default or survival. This simple way of binary description creates too strong a conditional default property. It is also associated with the simple way of specifying the correlation structure using only one parameter or pair-wise constant correlation in practice even though the original framework allows a completely flexible correlation matrix specification. Another possible reason is that this framework still misses certain fundamental driving factors such as volatilities of individual names in the framework.

We introduced the concept of excess loss distribution function, which allows us to examine most of the portfolio problems more intuitively from the graph. We also illustrated the method in pricing common basket derivatives of the i th-to-default contracts and CDOs, and examined the loss distribution of CDO tranches from an option perspective. Efficient algorithm for total loss distribution calculation using both recursion and conditional normal distribution, and detailed pricing formula for CDOs are provided in this overview.

We briefly discuss the risk measurement and risk management issues using the Gaussian copula function method. From the pricing formula we can obtain all necessary risk measures, such as spread DV01, jump-to-default risk, and gamma risk of individual spreads or the general index. We can also obtain the sensitivities of the portfolio of portfolio transactions with respect to each point of base correlation curve. The hedging instruments we could use include single name and index CDS, and index tranche, or even options on single name and index. However, we need to bear in mind whatever hedge strategies we have we still leave some residual risks. At different times we need to study and decide what residual risks we'd like to take. There are various criticisms about the framework and the need to come up with alternative models is pretty urgent. Many alternative methods and enhancements of the current ones have been suggested. However, the market hasn't agreed with a better replacement of the current model yet for basic CDOs and i th-to-default transactions. The ultimate judge of a model should be based on the hedging performance analysis of the model. Unfortunately we haven't seen extensive empirical studies about the performance of the model from hedging perspective. More empirical studies should be carried for any new model as well as the current Gaussian copula plus the base correlation approach.

The current copula framework gains its popularity due to its simplicity, and its capture of the most major driving factors in the framework. However, there is little theoretical justification of the current framework from financial economics. For example, to construct a joint risk neutral distribution simply using a copula function

for its individual risk neutral distribution is not necessarily correct to form a risk neutral joint survival time distribution. Its popularity with credit portfolio trading might match that of Black-Scholes formula for option valuation, but it lacks theoretical justification of replication as we see in Black-Scholes formula. We essentially have a credit portfolio model without a solid credit portfolio theory. More theoretical studies are needed to make further advancement in this area besides various extension and improvement which tries to fit the current market.

Reference

Finger, Chris, Conditional Approaches for CreditMetrics Portfolio Distributions, *CreditMetrics Monitor*, April 1999, pp.14 - 33.

Li, David X., On Default Correlation: A Copula Function Approach, *Journal of Fixed Income*, March 2000, pp. 41-50.

Li, David X. and Skarabot, Jure, Pricing and Hedging Synthetic CDOs, a chapter in the book "*Credit Derivatives: A Definitive Guide*" edited by Jon Gregory, Risk Publications, (2004).

Li, David and Liang, Michael, A Mixture Copula Function Approach to CDO and CDO Squared Pricing, (2005).

Andersen, Leif, and Sidenius, Jakob, All your hedges in one basket, *Risk*, November 2003.

Klugman, Stuart A., Harry H. Panjer and Gordon E. Willmot, *Loss Distribution: From Data to Decisions*, John Wiley and Sons, Inc., 1998.

Sathe, Y. S. and U. J. Dixit, On A Recurrence Relation for Order Statistics, *Statistics & Probability Letters*, 9, 1990.